

SciOdyssey.

A research layer over your agent harness.

Our agent can carry a research task to completion.

— PRESERVE THE RESEARCH WORLD · RESET THE RESEARCHER

ROBUST

SUPERVISABLE

BROAD SEARCH

SAME EVIDENCE
FRESH JUDGMENT
A WIDER SEARCH

Research
Runs
Longer.

01 / PROBLEM —

Why current agents fall short.

Sustained autonomous exploration encounters three critical bottlenecks:

01

Closed-world fragility

Rigid predefined pipelines shatter as soon as unexpected runtime errors emerge. In Cycle 1, a float32 JSON serialization bug halted logging; without autonomous self-healing, exploration dies permanently.

02

Single-trajectory inertia

Accumulating context history anchors the agent to past reasoning, trapping search in local basins (e.g., standard single-agent baselines stall at 0.6044 while endlessly fine-tuning hyperparameters).

03

Context inflation cost

Continually rereading deep conversational context costs 8× more tokens than the actual code edits, exhausting context windows and causing severe reasoning dilution.

TRADITIONAL AGENT HARNESS

Single Monolithic Context · Error Cascades · Token Waste · Local Optima

SCIODYSSEY RESEARCH LAYER

Ephemeral Fresh Scientist · Persistent Disk Memory · Verified Handoff

PARADIGM SHIFT

Decouple the **persistent research world** (disk memory) from ephemeral **researcher reasoning** (fresh context).

02 / SYSTEM —

One persistent world. Fresh scientific judgment.

Give the research world and the researcher different lifetimes.

THE DUAL-LIFETIME ENGINE

Context reset between trajectories · Audited persistence on disk

01

Environment

Task contract, data inputs, official evaluator & compute budget

→

02

Research World

Verified ledger, working priors, and reproducible code

→

03

Fresh Scientist

GPT ↔ Gemini rotation: clean hypotheses, code & experiments

→

04

META Review

Audit claims ↔ telemetry evidence; crystallize verified State

4-CYCLE MODEL RELAY

C1 GPT-5.6 (Baseline) → C2 Gemini-3.7 (Features) → C3 GPT-5.6 (Variance) → C4 Gemini-3.7 (Ensemble) → Gemini META throughout

Trajectory Handoff Loop:

Fresh Scientist reset per cycle; only audited empirical findings enter durable disk memory.

PERSISTENT RESEARCH WORLD

What survives across trajectories

A · VERIFIED EVIDENCE

Knowledge + Ledger

Complete metric trajectories, traceback failure logs, and full report artifacts stored in ledger.jsonl.

B · WORKING PRIORS

Brief + Hypotheses

Current conceptual understanding, domain beliefs, and exploration leads in brief.md, open to active revision.

C · REPRODUCIBLE STATE

Retained Code + State

Tested pipeline versions, serialized model weights, starter kit checkers, and full git commit provenance.

CODING HARNESS & RUNTIME

Physical Git worktree isolation · Sandbox evaluator · Structured JSON telemetry · Automated artifact capture

PARALLEL EXPLORATION

Independent branches explore in parallel → Post-hoc META review against validation split → Explicit state merge

The next Scientist inherits verified experience, not prior reasoning inertia.

03 / EVALUATION —

Five empirical claims. Measured evidence.

Standardized evaluation on KuaiRand-Pure benchmark · Fixed official evaluator · Complete telemetry accounting

01 MEASURED RESULT

The model got better.

0.6016000 → 0.6059363

Official FM baseline SciOdyssey submission

+0.0043363 Primary improvement

GAUC 0.6728421 (+0.0054) nDCG@5 0.5390304 (+0.0033) LogLoss 0.38102 (-0.0032)

46 categorical interaction features · 8-seed Factorization Machine ensemble · 100% CPU / NumPy training.

02 DIRECT BENCHMARK CONTROL

Beat Codex on this task.

0.6044533 → 0.6059363

Direct Codex Goal control SciOdyssey retained

+0.0014830 Primary advantage

Direct single-agent baselines anchored in local parameter tuning across 22 experiments; SciOdyssey broke out into superior feature space via world resets.

03 SUSTAINED SEARCH

Systematic multi-cycle progress.

0.6016310 C1: Valid baseline

0.6040901 C2: Feature expansion

0.6044289 C3: Variance reduction

0.6059363 C4: 8-seed FM recipe

4 autonomous cycles 13 named experiments 7 full evaluations

Monotonic validation progression across distinct cycles; breakthroughs driven by mechanism changes rather than a single lucky edit.

04 AUTONOMOUS RECOVERY

Our agent is robust.

01 · PRESERVE

NumPy float32 broke JSON serialization in Cycle 1; logs stayed intact.

→

02 · REPAIR

Scientist diagnosed traceback and patched scalar serialization helper.

→

03 · RETAIN

Rerun succeeded; META validated proxy fidelity & preserved State.

Zero human debugging intervention; autonomous traceback self-healing in Cycle 1. META audits proxy fidelity and rejects spurious state promotions.

05 RESOURCE ACCOUNTING

We are not tokenmaxxing.

TOTAL PROJECT USAGE

~\$10

subscription-equivalent estimate

HARDWARE

1 × MacBook M2

All training on 1 laptop · 0 GPU-hours

SUBSCRIPTIONS

GPT Plus + Gemini Pro

7 days continuous exploration

TOTAL LLM TOKENS

48.24M

4.02M non-cache · 8.5× token reduction

100% Autonomous execution (0 manual code interventions)

170,588 rows checked by Starter Kit alignment checker

0 GPU-hours (NumPy / CPU training)

| All results verified on KuaiRand-Pure official validation split with complete reproducible artifacts.

04 / INSIGHTS —

What the run taught us.

01 · MODEL DIVERSITY

Depth + Breadth Synergy.

GPT 5.6 Luna (depth: refine ideas) + Gemini 3.7 Flash (breadth: open new directions) in one shared world pushed score from 0.6052 (Gemini-only) to 0.6059363 (GPT+Gemini).

Gemini-only 0.6052 → GPT+Gemini 0.6059363

02 · PARALLEL SYNTHESIS

Cross-Branch Idea Transfer.

Reviewer synthesized winning Branch B (0.6048) with modeling ideas from lower-scoring Branch C (0.6039) for pairwise ranking and rank normalization, yielding 0.6055536.

Branch B+C Ideas → Synthesis 0.6055536

03 · UNATTENDED EXECUTION

Branch Isolation on Actions.

Git worktrees isolate state to prevent collisions; GitHub Actions runs, codes, and commits overnight without human intervention, enabling continuous unattended progress.

Git Worktree Isolation · CI/CD Automation

05 / EXPERIENCE —

Start your journey.

task.md

What to solve

Formal task contract: research objectives, dataset inputs, compute constraints, and official evaluation metric protocol.

PERSONAL.md

How to work

User research preferences, model guidance, exploration style, tool permissions, and supervisory checkpoints.

SUPERVISION TIERS

Tier 1 Unattended Overnight Runs · Tier 2 Checkpoint Review (META) · Tier 3 Steer Working Priors

1. Define task.md → 2. Single-step test → 3. Autonomous search → 4. Parallel explore → 5. Inspect evidence

DECLARATIVE PROTOCOL

Human defines boundary and preferences (task.md + PERSONAL.md); SciOdyssey autonomously iterates, heals, and crystallizes verified evidence.

Autonomous by default.

Supervisable by design.

Coding agent harness · Execution session

NATURAL LANGUAGE ENTRY

> Use the research-agent skill. Follow task.md and PERSONAL.md. Run autonomously.

✓ [Environment] Loaded KuaiRand-Pure dataset (170,588 rows) · Verified official evaluator

✓ [Cycle 1 · GPT-5.6] Baseline verified: Primary 0.6016310 · Self-healed float32 JSON serialization

✓ [Cycle 2 · Gemini] Feature expansion: 46 user-video interaction features · Primary 0.6040901

✓ [Cycle 3 · GPT-5.6] Variance reduction & seed averaging · Primary 0.6044289

✓ [Cycle 4 · Gemini] 8-seed Factorization Machine ensemble · Primary 0.6059363 (+0.0043363)

✓ [META Review] Verified claims ↔ telemetry; crystallized reproducible State to durable disk

Status: Completed

Duration: 7 days unattended

Compute: 1x MacBook M2 (0 GPU-hr)

Tokens: 48.24M

github.com/zc6600/research-agent-kuairand ↗

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Sources: slides/slides.md · docs/ARCHITECTURE.md · docs/FINAL_REPORT.md · KuaiRand-Pure submission ledger

ROBUST · SUPERVISABLE · BROAD SEARCH
A Persistent Research World for Autonomous Science